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ORIGINAL RESEARCH

A Novel Machine learning and Internet of Things (IoT) Based Dual Monitoring System for Proactive Botnet Attack Prevention

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Abstract

Existing techniques in anti-botnet security are focused primarily on detection and analysis or mitigation. However less attention is on botnet attack prevention, which is needed for sectors like healthcare, defence or nuclear. Prevention is must for critical sectors as detection or analysis measures commence after the botnet damage (as services becomes unavailable) which is postmortem analysis. The current study aims to improve anti-botnet IoT network security by preventing botnet formation, which is critical for robust IoT applications. A proposed dual monitoring system is introduced to detect malicious behaviour at both the device and network levels, utilising rule-engine based analysis and machine learning techniques. Early detection allows for more timely preventive measures, such as isolating compromised devices before they are integrated into botnets. At the device level, in this research rule engine is used to define and monitor individual rules for each IoT device. For network-level detection, logistic regression was used to classify network traffic and exploratory data analysis to identify unusual scanning activity. The defined method increases the prospect of detecting malicious behaviour while decreasing the likelihood of botnet formation by combining device and network-level monitoring. As a precaution, devices that are flagged as malicious at both levels are isolated. Testing in an IoT-based healthcare network shows that the dual monitoring approach improves malicious behaviour detection and botnet threat mitigation. When compared to existing literature, the proposed dual monitoring system shows a marked improvement (15% higher on accurate detection, 20% higher on recall, 10% higher on traffic anomaly detection and 25% higher on identifying compromised devices) in the identification and mitigation of potential botnet activities within an IoT network.

Keywords Machine learning · Internet of things · Botnet attack · Predictive security · Proactive security · Preventive security

Introduction

The combination of IoT devices and machine learning has accelerated consumer and Industrial IoT growth. Although IoT has impacted every field, we need better recognition of security threats in IoT. According to a 2022 Gartner report, 90% of IoT devices are vulnerable to attacks, emphasizing the critical need for comprehensive security policies and industry-wide standards. Many researchers have stated that threats in IoT should be addressed proactively in collaboration with research communities and manufacturers of IoT devices [20, 33, 40, 41]. Major concerns of IoT users are about privacy and security as attackers can steal personal information and monitor all user activities [32, 49].

Understanding IoT attack vectors and their impact is critical [37]. IoT network attacks include jamming, sinkhole attacks, node tampering, black hole attacks, reply attacks, wormhole attacks, Sybil attacks, selective forward attacks, hello flood attacks, man-in-the-middle attacks, reply attacks, sleep deprivation attacks, DOS attacks, malicious node injection, and botnet attacks[6, 10]. The majority of the attacks can potentially compromise any one of the three components of the confidentiality, availability, and integrity (CIA) triad. However, the highest threat is from the botnet

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attack as it can affect availability, confidentiality, and availability all at the same time[2].

A botnet is defined as a network of a bot. A bot being a machine that searches for the most vulnerable device, infects it, and converts it into a slave. Reportedly, the first botnet emerged in 1988, while the first IoT botnet emerged in 2013 [13, 42]. The IoT botnets uses devices that are often located remotely. These devices have minimal user interaction, allowing IoT botnets to remain undetected for extended periods. IoT device manufacturers prioritize cost over security, resulting in widespread deployment of IoT networks with default, easily exploitable credentials. The lack of built-in security measures in IoT devices increases vulnerability, making these devices attractive targets for attackers[9]. All of this implies that the IoT botnet concern should be addressed proactively.

The existing research on IoT botnets is categorized into four main areas: analysis, detection, mitigation, and prevention. In the analysis domain, researchers employ honeypots[25, 31, 39, 47, 50], sandboxes [7, 23], testbeds [5, 22]?, and datasets[19, 28, 29] to understand botnet characteristics and behaviors. Various honeypot types are proposed, and network flow analysis is highlighted for effective attack detection. IoT Botnet detection methods, explored in the second category, involve algorithmic and machine learning models, such as the IoT-BDA framework [45] and Edge2Guard [43]. Hybrid approaches and byte sequences from executable files are also utilized for detection. Machine learning [4, 14, 24, 27, 35] and deep learning [11, 16, 26] techniques are extensively employed, with studies comparing various algorithms and proposing ensemble methods for model selection[12]. The third category focuses on IoT botnet mitigation strategies post-detection, such as isolating infected devices and applying security patches. The white hat worm [36], SDN control panels [21, 48], and Dytokinesis are proposed for efficient mitigation. Dytokinesis is a mitigation technique inspired from the biological process of cytokinesis. In the same way, Dytokinesis splits datasets promptly and precisely. It works in stages, dividing the dataset into normal and anomalous categories using Gaussian kernel techniques and Empirical Data Analysis (EDA) [34]. Lastly, in the prevention category, researchers advocate proactive measures to prevent IoT botnets. Strategies include secure IoT architectures[3, 30], strong authentication, and best practices. Several frameworks[44] and models [8], incorporating machine learning and deep learning, are proposed for preventive solutions[15, 17]. Researchers proposed hardware-based solutions to prevent botnet attacks at the device level[18, 38, 46]. The literature review concludes that, majority of research efforts are in detection and mitigation, whereas work on prevention is limited, comprising only 15% of the overall research. Hence, the current research emphasises the importance of prioritizing prevention to reduce the impact of IoT botnet attacks. The statistical summary in Fig. 1 provides an overview of the distribution of research efforts across the four categories.

According to the findings, detecting and mitigating botnets is time-consuming, and efforts toward IoT botnet prevention are limited. The loss caused by IoT botnet attacks can be reduced by preventing the formation of IoT botnets rather than detection or mitigation. The proposed work's goal is to propose and investigate a proactive security strategy for protecting IoT networks from botnet attacks. The following are the main contributions of the work.

1. Conceptualized a dual monitoring system to prevent bot formation in an IoT system.
2. Designed and developed dual monitoring to prevent bot formation in IoT systems.
3. Deployment of the developed model on the healthcare system.
4. Real-time device-level monitoring: Utilizing the Things board's rule engine to monitor and identify malicious behavior at the individual device level.
5. Profiling and analyzing existing datasets to build a machine-learning model.
6. Developed a dataset for predicting likely bots at an early stage by identifying malicious behavior of IoT devices.

The paper is organized as follows: Section 2 gives an overview of the dual monitoring system. Section 3 covers System implementation and key considerations, while Section 4 presents experimental results from an IoT-based healthcare network. Section 5 discusses the approach of validating the proposed system, and the conclusion summarizes findings, discusses future improvements, and highlights the research's significance in enhancing IoT network security.

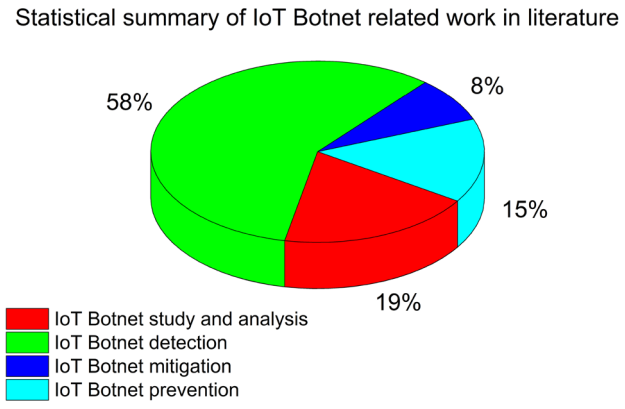


Fig. 1 Statistical summary of IoT botnet-related work done in literature

Methodology

The dual monitoring system depicted in Fig. 2 is designed to detect and prevent malicious behavior in IoT devices at both the device and the network level. The system follows the monitor-identify-isolate principle to detect potential threats early in the botnet cycle and prevent IoT devices from being compromised and turned into bots.

Overview of Dual Monitoring System

The system consists of two main modules: device-level monitoring and network-level monitoring. Further network monitoring is performed through ML and data exploratory approaches.

Device-level monitoring focuses on closely monitoring the activities of each IoT device in the network to identify possible threats. It includes three sub-modules: framing rules, monitoring devices, and scanning database.

The framing rules sub-module defines specific rules for each IoT device based on its limited functionality and predictable behaviour. These rules consider device status, access levels, data generation rate, and the range of measured sensor values. The rules are defined and monitored using a rule engine like ThingsBoard, an open-source IoT platform. If a device violates these rules during the monitoring phase, an alarm is generated and stored in the database for further analysis.

The monitoring devices and the scanning database submodule test the device-level functionality by deploying various scenarios using curl commands. If an IoT device's behaviour does not follow the established rules, the system generates warnings and stores them in the database. Frequent rule violations are considered potential malicious behaviour, indicating IoT devices can be turned into bots in the future. The scanning activity results in a list of devices that have crossed the defined rule violation count threshold or the average rule violation count of the network. The network-level monitoring module focuses on analyzing network traffic to detect malicious behavior and unauthorized access.

Features with discriminative power are specific network traffic characteristics employed to detect anomalous behaviour and scanning activity. Examples include bytes per packet and flow per hour, distinguishing between typical and atypical patterns. Examining these characteristics makes it easier to spot deviations from the usual pattern and identify possible dangers. These metrics are essential to identify anomalies in network traffic. In this research, network traffic characteristics, such as bytes per packet, flow per hour, and other network traffic characteristics elaborated further are used to identify abnormal behavior and scanning activities. The network level module includes three steps: building a machine learning (ML) model, generating traffic, and testing and analyzing the traffic.

While building the ML model, two datasets are selected: a realistic IoT botnet dataset consisting of legitimate and simulated IoT network traffic and the Cyber Threats and Underground (CTU-13) dataset, which includes 13 botnet

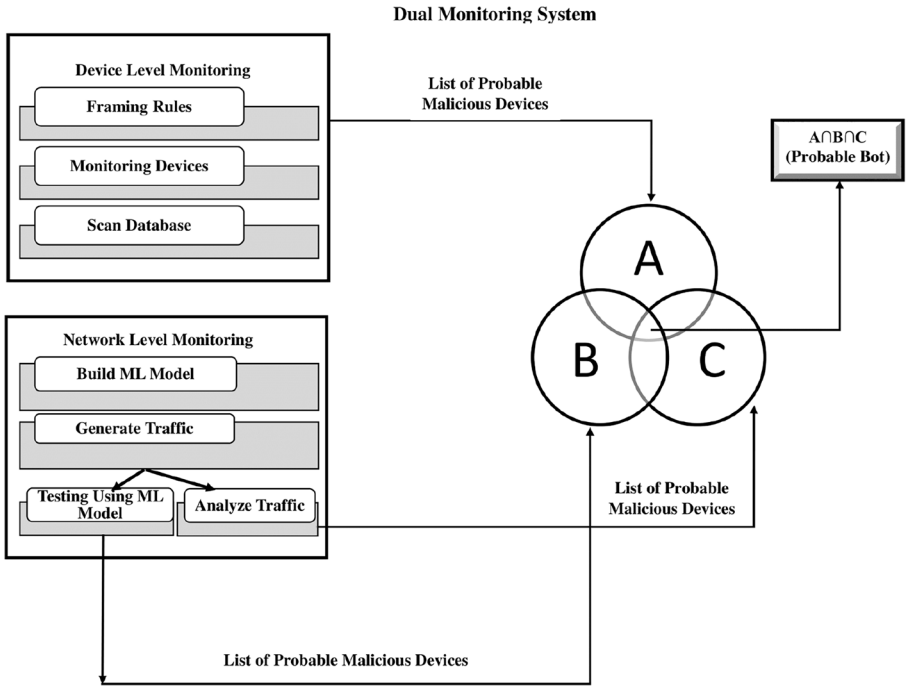


Fig. 2 Block diagram of dual monitoring system

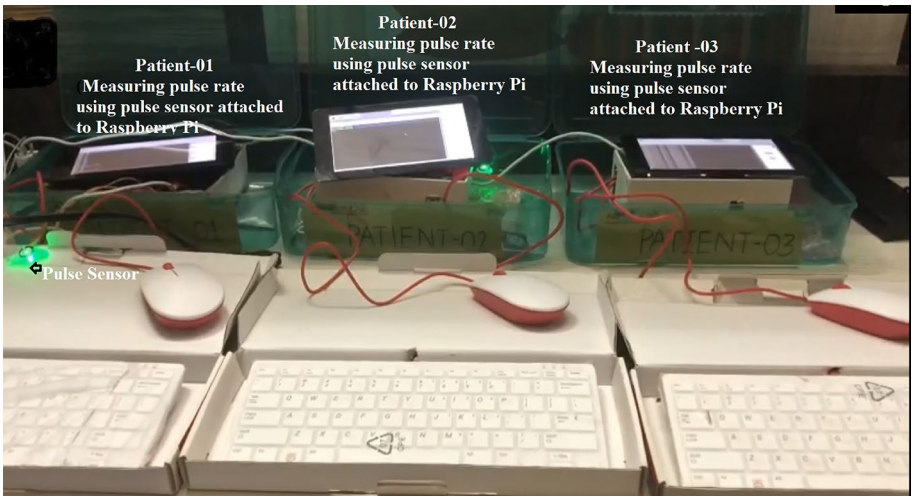


Fig. 3 Actual setup of IoT healthcare network

traffic scenarios. Various ML algorithms, such as logistic regression, support vector machine (SVM), and Naive Bayes, are evaluated for classifying network traffic into benign and malicious. The best algorithm is chosen based on accuracy, precision, recall, f1-score, and ROC curve.

In the traffic generation step, benign and malicious traffic were generated using an IoT-based small healthcare monitoring setup. The malicious traffic was generated through active probing attacks and rule violations at the device level. Wireshark was used to capture the traffic in pcap format, while flow data was generated using the ARGUS tool to enhance detection accuracy.

Finally, the generated traffic is tested and analyzed using the built ML model. Traffic classified as malicious by the ML model is added to the set of malicious devices. The system also captures behavioural changes in IoT devices by analyzing benign traffic. Devices that generate more traffic and exhibit abnormal behaviour are added to the set of potentially malicious traffic.

The device list, obtained by taking the intersection of all three groups (A, B and C) consists of devices that are highly likely to turn into bots in the future. The groups are defined as:

1. Group A: probable malicious devices identified using device level monitoring,
2. Group B: probable malicious devices identified using network level monitoring through ML approach,
3. Group C: probable malicious devices identified using network level monitoring though data exploratory approach,

These devices are isolated from the IoT network for further analysis, aiming to prevent the emergence of a botnet in the network.

Overall, the system follows a monitor-identify-isolate principle to detect and prevent IoT devices from being compromised and forming a botnet. It achieves this by continuously monitoring the behaviour of devices at both the device and network levels. This proactive approach allows for the early identification of potential threats, enabling prompt action to isolate devices exhibiting malicious behaviour from the network. Once isolated, further analysis and preventive measures can be implemented to safeguard the overall security of the IoT ecosystem.

System Implementation and Key Considerations

In this section, we delve into the details of implementing the proposed system for enhancing IoT network security. The system employs a dual monitoring approach, focusing on device and network levels to detect and prevent potential botnet attacks. Key considerations will be discussed, including setting up a healthcare monitoring IoT network, configuring the ThingsBoard platform, establishing rules using a centralized rule engine, and integrating machine learning for traffic analysis. Additionally, the section outlines the significance of defining thresholds and scanning intervals and generating traffic datasets for both benign and malicious activities. These implementation steps play a crucial role in the system's ability to identify and mitigate security threats within an IoT environment proactively.

Setting up IoT-Healthcare Network

Due to its superior speed and processing power, the deployment process begins with establishing a small-scale healthcare monitoring IoT network using a Raspberry Pi 4 GB RAM. The network uses wireless and Bluetooth capabilities

to measure pulse values using a pulse sensor requiring 3V or 5V.

Figure 3 illustrates the actual setup, providing a visual representation of the physical arrangement of components. In the use case for the healthcare system, the proposed dual monitoring system is designed for deployment in IoT networks with customizable rule engines. The healthcare sector is chosen for deployment due to its critical nature, potential transformative impact on patient care, and the need for efficient remote monitoring. The system addresses challenges in traditional healthcare systems, especially during events like the COVID-19 pandemic, offering real-time monitoring for timely interventions. Its scalability and versatility make it adaptable to various healthcare settings, with the integration leveraging existing IoT-enabled medical devices. Acknowledging security concerns, the researchers emphasize the need for robust measures to ensure data integrity and patient confidentiality in healthcare monitoring systems. The chosen healthcare domain reflects the system's potential to improve patient care, remote monitoring capabilities, and overall healthcare outcomes.

Configuring the ThingsBoard Platform for Rule Enforcement

Due to limited input and output capabilities, IoT devices exhibit predictable behaviour. For instance, a temperature sensor in a smart AC provides numerical values within a defined range. This distinct and predefined set of capabilities allows for rule definition for each device, encompassing both device-specific and generic rules such as device profile, status, and access levels. Tailoring rules to accommodate variations in data generation rate and measurement range specific to each device ensures effective rule enforcement in IoT networks.

ThingsBoard, an open-source IoT platform, the system utilizes its rule engine for defining and monitoring rules [1]. The event-based workflow of ThingsBoard facilitates effective device management, data collection, processing, and visualization. All the rules are consolidated and validated by the comprehensive rule engine, as illustrated in Fig. 4.

The centralized rule engine is a pivotal component in the system, playing a crucial role in enforcing and applying all defined rules to incoming data. Operating as a centralized decision-making entity, it processes and evaluates the rules against the data, making informed decisions to enhance the security of the IoT network. The proposed system establishes four rules for the meticulous monitoring of individual IoT devices in the network, as given below.

1. Device status,
2. Device activity,
3. Data range validity,
4. Network association

Establishing Thresholds and Scanning Intervals for Security Monitoring

To gauge the overall security posture of the defined network, the system computes the average number of rule violations observed across all devices. This value is a crucial metric in determining the network's security status. However, setting a threshold for this average value is problem-dependent and requires careful consideration.

The proposed solution adopts a probabilistic approach to detect potentially malicious machine behaviour. In line with this approach, a threshold value of 50% is chosen as the system's threshold. This threshold aligns with the commonly used decision threshold of 0.5 in logistic regression, a ML technique employed in the system. By employing this

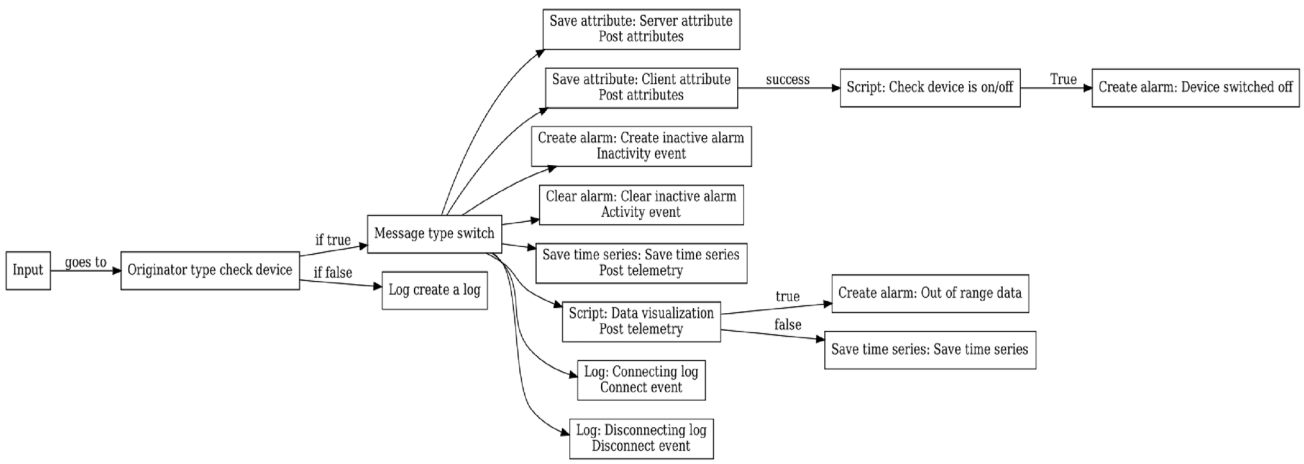


Fig. 4 Overall rule engine

threshold, the system classifies the device's behaviour as potentially malicious (above the threshold) or non-malicious (below the threshold). It aids in identifying devices that may be susceptible to becoming part of a botnet, enabling administrators to take proactive measures to prevent such occurrences and ensure the network's overall security.

Under the monitoring and scanning phase, the network is scanned after interval 't' to identify the malicious behavior of the IoT device. The effectiveness of the dual monitoring system in identifying potentially malicious IoT devices relies on the scanning interval for the warning table. The choice of scanning period is contingent on the IoT network's size and the desired security level. In smaller networks, shorter scanning intervals are advantageous due to fewer devices, allowing for swift identification of rule violations and immediate action against potential threats. Shorter scanning intervals enhances overall network security. In larger networks with increased computational intensity, shorter intervals remain desirable as they enable a higher prevention approach, promptly detecting and responding to rule violations. Despite the computational cost, which refers to the resources and time required to perform the scanning and analysis operations on the network, minimizing the time between scans is crucial to quickly isolate potentially malicious IoT devices, reducing the risk of security breaches. The selection of the scanning interval involves balancing computational efficiency with the required security level, considering the network's size, complexity, and criticality. Tailoring the scanning interval to the network's characteristics allows the dual monitoring system to effectively identify and mitigate potential security threats, maintaining a secure environment.

Table 2 Performance of the models during training built using IoT-BoT Dataset

Metrics	Logistic regression	SVM	Naive Bayes classifier
Accuracy	0.9971	0.9997	0.9970
Precision	0.9971	0.9998	0.999
Recall	1.0	0.9999	0.9970
f1score	1.0	1.0	1
Training time (ms)	44.64	2400	9.998

Building ML Models for Traffic Analysis

When creating an ML model, two things are critical. The first is selection of datasets for training the model, and the second is selection of ML algorithms for modeling.

Selection of Dataset: The efficiency of an ML model depends on the quality of the dataset used for training. For this purpose, two distinct datasets are selected to build the ML model. The first dataset is the BoT-IoT dataset, created in 2018 in the Cyber Range Lab of UNSW Canberra, representing a realistic IoT network with both legitimate and simulated botnet traffic. Available in pcap, argus, and CSV formats, the dataset is approximately 16.7 GB and includes various fields like protocol, source and destination addresses, number of packets sent, data present in each packet in bytes, and more. A subset of the data featuring the top ten characteristics chosen through statistical analysis is used for optimal results (table 1). Notably, this dataset is considered the first realistic attack dataset in an IoT environment. The second dataset is the CTU-13 dataset, generated in 2011 at CRU University in the Czech Republic, encompassing 13 botnet traffic scenarios, both benign and malicious. Scenario 8 from 13 botnet traffic scenarios, involving Internet Relay Chat (IRC) for network scanning, is specifically chosen for developing the machine-learning model in the proposed system.

Table 1 Featured used for building the ML model

SNo	Feature	Description
1	srate	Source_to_destination packets per seconds
2	drate	Destination_to_source packets per seconds
3	max	Maximum duration of aggregated records
4	state_number	Numerical representation of feature state (specific attribute within the dataset that represents the state of a network connection)
5	mean	Average duration of aggregated records
6	min	Minimum duration of aggregated records
7	stddev	Standard deviation of aggregated records
8	seq	Argus sequence number
9	N_IN_conn _srcIP	Number of inbound connections per source IP
10	N_IN_conn _dstIP	Number of inbound connections per destination IP

Table 1 details the fields utilized during the development of the ML model, combining features from both datasets.

Table 2 provides a comparative analysis of the performance metrics for the three different ML models built on the CTU-13 dataset, scenario number 8.

Selection of ML algorithm: Classifying network traffic involves a binary classification problem, distinguishing between benign and malicious classes. Supervised machine learning offers various algorithms, with logistic regression, support vector machine (SVM), and Naive Bayes being the top contenders.

In the training phase, six ML models were built using two different datasets: IoT-BoT and CTU-13. Three models were trained on each dataset, resulting in a diverse set of models for evaluation and comparison. The IoT-BoT dataset was used to train three ML models, while the CTU-13 dataset was used to train the other three models. Using multiple datasets, the models can capture different patterns and behaviours in each dataset, potentially leading to more robust and accurate results. Training ML models on different datasets helps test their performance on varied data sources, allowing for a better understanding of their generalizability and effectiveness in different scenarios.

Using multiple datasets in the training phase provides a comprehensive and diverse foundation for building ML models, enhancing their ability to handle different data types and improve overall performance.

Table 3 presents the detailed performance results of the models during training, specifically focusing on those built using the CTU-13 dataset.

The performance of these algorithms is assessed using different evaluation metrics, including accuracy, precision, recall, f1-score, and the receiver operating characteristic curve (ROC curve). The most suitable algorithm for ML model development is selected based on a comprehensive analysis of these metrics.

When comparing Tables 2 and 3, it is observed that the ML model built with logistic regression demonstrates good results for both the IoT-BoT and CTU-13 datasets. Additionally, logistic regression exhibits a shorter training time compared to the support vector machine (SVM) algorithm.

Table 3 Performance of the three different ML models build on CTU-13 dataset

Metrics	Logistic regression	SVM	Naive Bayes classifier
Accuracy	0.9667	0.8107	0.7224
Precision	0.9387	0.84968	0.6430
Recall	0.9986	0.7550	0.9998
f1score	0.97	0.78	0.78
Training Time(ms)	4.33	8.76	7.3963

Considering the aim of the proposed system to provide a simple and cost-effective solution, the choice of logistic regression as the classification algorithm for building an ML model is justified. Logistic regression is known for its simplicity and efficiency, making it a suitable choice for classifying generated traffic.

Generating Benign and Malicious Traffic for Evaluation

The objective of traffic generation in the experimental setup is twofold: creating datasets for benign IoT traffic and malicious network traffic. Generating benign traffic is crucial for understanding normal IoT device behaviour within a network, facilitating the detection of deviations caused by malware infections or attacks. Malicious traffic is generated through an active probing attack, employing Nmap for port scanning and OS fingerprinting, coupled with rule violations using curl commands at the device level. These rule violations induce abnormal behaviour captured in the traffic. Wireshark captures traffic in pcap format, which is later processed by the Audit Record Generation and Utilization System (ARGUS) tool to generate flow data with additional features enhancing ML classification model accuracy. All generated traffic files are then stored in the database for further analysis. Tables 4 and 5 show the statistics of the generated traffic.

Experimental Evaluation & Result and Discussion

This section deals with a discussion of the experimental evaluation of the dual monitoring system level-wise followed by an integrated result discussion.

Table 4 Summary of the Benign Traffic generation scenario

Device Details	no. of devices	hrs	no_of packets	pcap size
Raspberry Pi measuring heartbeat	10	6	35,89,105	1389 MB

Table 5 Summary of the malicious traffic generation scenario

Device Details	no of devices	hrs	no. of packets	pcap size
Raspberry Pi measuring heartbeat	10	1.5	176,819	22.5MB

Table 6 Statistical summary of the alarm generated by the IoT devices in the network

No. of IoT devices in the network	10	
No. of alarm generated at time t	63	
Initial Threshold Value (th)	50 %	
Average No. of alarm generation	6.3	
Device	Count of alarm	Device_wise %of alarm generation
D1 (patient 1)	23	36.50 %
D2 (patient 2)	04	6.3 %
D3 (patient 3)	18	28.57 %
D4 (patient 4)	05	7.93 %
D5 (patient 5)	04	6.3 %
D6 (patient 6)	03	4.76 %
D7 (patient 7)	03	4.76 %
D8 (patient 8)	02	3.17 %
D9 (patient 9)	01	1.58 %
D10 (patient 10)	00	0 %

Device-Level Monitoring

Table 6 displays a statistical summary of the observations made during the scanning phase in device-level monitoring of the experimental setup. As the network is small in size the chosen time interval is small to ensure more security (Table 6).

A setup of 10 IoT devices was created, with randomly generated traffic and malicious activity. The total number of alarms generated in the network divided by the total number of devices yields the average number of alarms. This average number of alarms is used as a parameter to determine whether a device is potentially malicious. An IoT device is deemed malicious if its average alarm count is higher than 50%.

Based on the scanning phase of the device-level monitoring, it has been found that Device D1 (Patient 1) and Device D3 (Patient 2) have generated a higher number of alarms compared to the average number in the system at the given time t. This observation suggests that these devices have elevated alarm activity levels, which could indicate a potential anomaly or violation of predefined rules.

It is essential to investigate these devices further to understand the reasons behind the increased alarm count and take appropriate actions to address potential issues or security threats. So, at the device-level monitoring stage, the output obtained is Set A = D1 (Patient 1), D3 (Patient 3), as shown in Fig. 5.

It is crucial to address the behavior of these devices promptly to ensure the overall security and integrity of the IoT network.

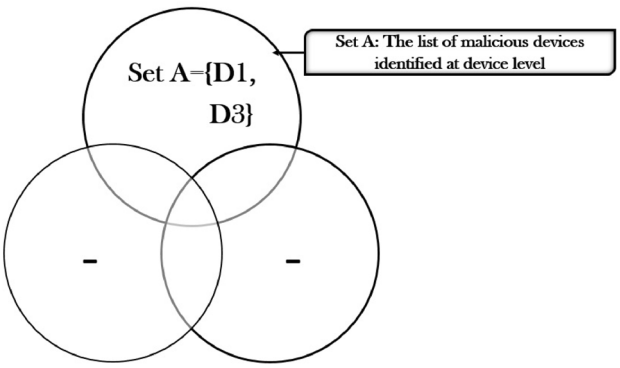


Fig. 5 Venn diagram: depicting the output of device level monitoring stage only

Network Level Monitoring

Traffic Analysis Using Machine Learning Model and Inference

The generated traffic from the setup is tested using the logistic regression-based pre-build ML models.

The statistics in Figs. 6 and 7 indicate that the models have low bias and variance. Low bias suggests that the models can accurately capture the underlying patterns and relationships in the data. In contrast, low variance indicates that the models are relatively sensitive to small fluctuations in the input data. Using logistic regression-based ML models in the proposed system offers a simple and efficient solution for classifying generated traffic while maintaining good performance with low bias and variance.

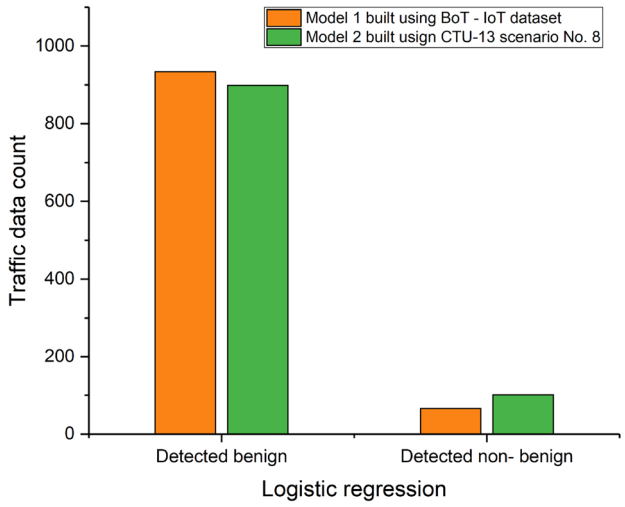


Fig. 6 Statistics of predictions of benign traffic using Logistic Regression of both models

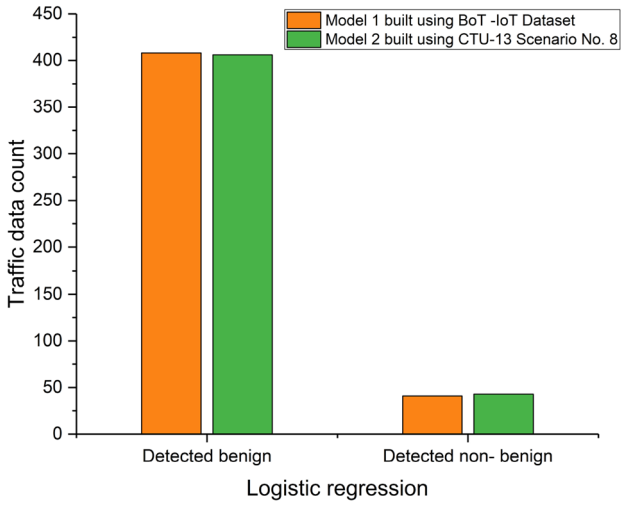


Fig. 7 Statistics of predictions of malicious traffic using Logistic Regression of both models

Table 7 Performance of the Model 1 and Model 2 during testing of generated traffic

Metrics	Model 1	Model 2
Accuracy	0.92	0.90
Precision	0.86	0.80
Recall	0.90	0.90

Observing Table 7, it can be concluded that both models, built with logistic regression, exhibit reasonable recall rates. Recall, also known as sensitivity or true positive rate, measures the ability of a model to identify positive instances correctly.

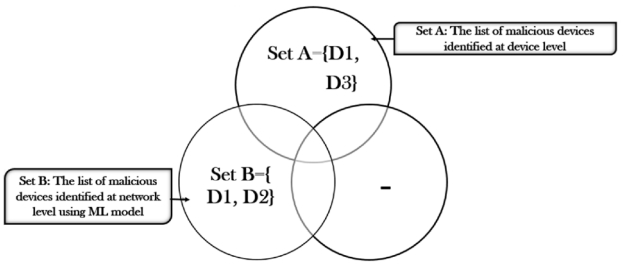


Fig. 8 Venn diagram: Depicting the output of device level and network-level monitoring stage using ML model

In the case of the logistic regression model, it achieves a recall rate of more than 0.90 in both models, indicating that it correctly identifies more than 90% of the positive instances in the testing dataset.

In the network traffic analysis scenario, achieving a high recall rate is crucial to ensure that all potentially malicious traffic is correctly classified as malicious. It indicates the model's effectiveness in accurately detecting and identifying malicious behaviour. Based on the results obtained from both models built using different datasets, it is observed that they achieved a 90% recall rate. This 90% recall rate implies that the models successfully identified 90% of the potentially malicious traffic, correctly classifying it as malicious. Overall, the consistent 90% recall rate achieved by both models suggests their effectiveness in detecting and identifying potentially malicious traffic in the IoT network, providing valuable insights for enhancing security and ensuring early detection of malicious behaviour.

The IP addresses are traced for the traffic that is classified as malicious by the ML model. It has been determined that devices D1 (Patient 1) and D2 (Patient 2) are identified as malicious traffic sources. These devices have exhibited behaviour indicative of potential bots within the network. As a result, Set B, which includes D1 (Patient 1) and D2 (Patient 2), is generated as the output of the network-level monitoring stage using the ML method.

Figure 8 visually represents the output, highlighting devices D1 (Patient 1) and D2 (Patient 2) as probable bots within the network. This identification is based on their involvement in generating malicious traffic, as detected and traced through the IP addresses associated with the traffic dataset.

The network-level monitoring stage, employing the ML model, plays a crucial role in identifying and flagging potential bots within the IoT network. By accurately identifying devices D1 (Patient 1) and D2 (Patient 2) as probable bots, the system enables proactive measures to be taken for further investigation, isolation, and mitigation of their malicious activities.

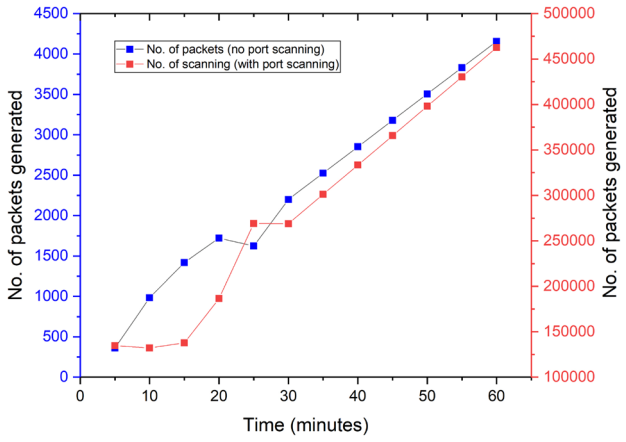


Fig. 9 Comparison of traffic statistics in the presence and absence of port scanning activity

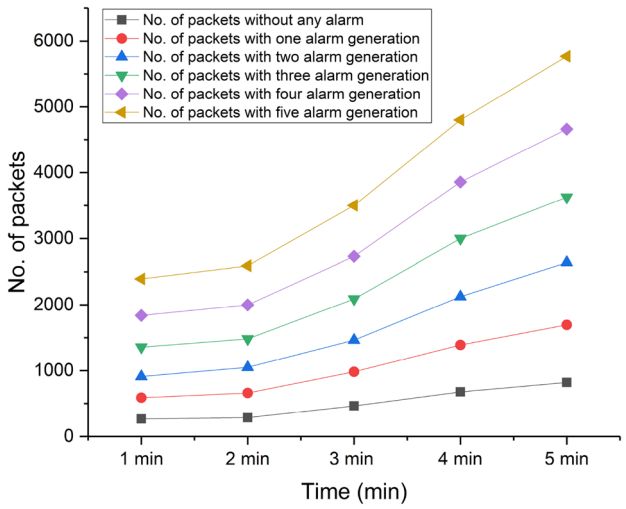


Fig. 10 Comparison of the number of packets generated with and without an alarm

Traffic Analysis Using Data Analysis Technique

The second method of analyzing patterns from network traffic involves the use of exploratory data analysis techniques. In this method, the traffic volume generated during benign scenarios is compared with the traffic volume generated during scanning activities. This allows for a comparison of network traffic statistics in the presence and absence of scanning activity.

Figure 9 visually represents the comparison of network traffic statistics. It provides insights into the variations in traffic volume observed during scanning activities compared to the traffic volume generated in benign scenarios. Analyzing these patterns highlights that scanning activities

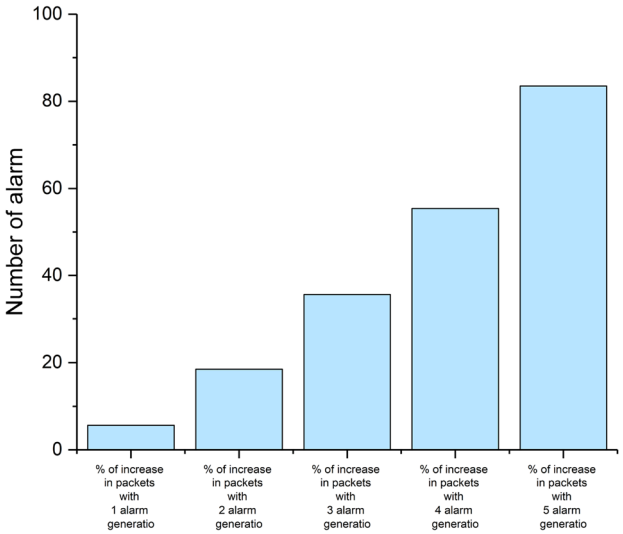


Fig. 11 Overhead cause due to alarm handling

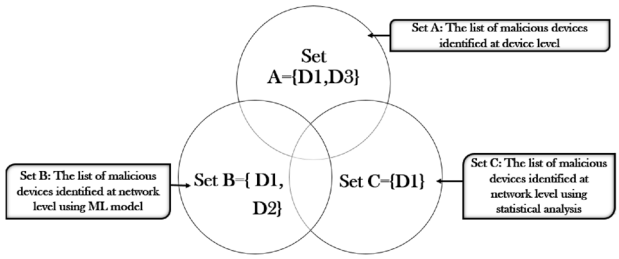


Fig. 12 Venn diagram: depicting the output of device level and network level using ML and analytics methods

generate more traffic-this sudden increase in traffic aids in identifying anomalies associated with scanning activities.

Similarly, device-level alarms triggered by rule violations also contribute to increased network traffic. The server responds to each alarm by sending three to five extra packets, resulting in a higher number of packets being generated.

Figure 10 compares the number of packets generated with and without the alarm scenario, highlighting the significant increase in packet count when alarms are present. This increase in traffic and associated overhead, shown in Fig. 11, highlights the imposed additional load on the system.

The analysis of the experimental results indicates that Device D1 (Patient 1) exhibits abnormal behaviour in generating a higher number of packets due to increased alarm generation.

Figure 12, shows the output of the analysis, indicating that Device D1 is flagged as a potentially compromised device or a victim of an active probing attack. This information is crucial for taking proactive security measures, such as isolating the device, investigating the attack source, and

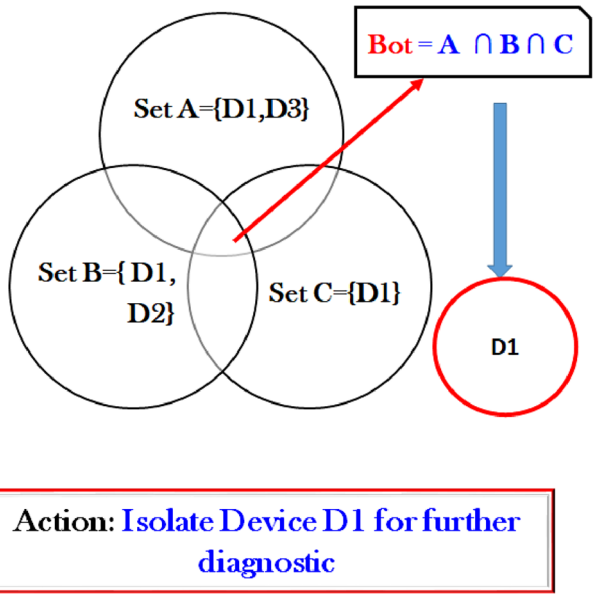


Fig. 13 Venn diagram: depicting the final output of dual monitoring system

implementing appropriate security controls to prevent further malicious activities.

Integration of Result

Figure 13 provides a graphical representation of the final outcome, where IoT device D1 is isolated due to its presence in all three sets A, B, and C, indicating its high likelihood of being a bot or compromised device.

Isolating compromised IoT devices within a network is a pivotal strategy for preventing the escalation of malicious activities. This approach serves multiple purposes, from preventing further harm to conducting focused examinations of compromised devices. Swift isolation enables a thorough investigation, determining the extent of compromise, identifying vulnerabilities or malware, and implementing targeted security measures.

By promptly isolating compromised devices, the proposed dual monitoring system ensures the overall security of the IoT network. This proactive measure safeguards other connected devices, preserving the integrity and privacy of network data.

Isolating identified malicious devices in the early botnet lifecycle effectively obstructs the progression of botnet formation. This proactive step serves as a deterrent against potential threats, contributing to a secure IoT environment.

Proactively isolating potentially malicious IoT devices ensures they are unable to execute harmful activities, maintaining the security and integrity of the IoT network. Early

Table 8 Level-wise significance, weight, weightage, and output

Level	Significance of level and method	Weight	Weightage	Rationale	Level-wise Output
Device Level (A)	Close monitoring makes it easier to capture all behavioral changes. The Thingsboard Rule Engine is used to deploy the rules	100	30%	Root level detection. The closest layer helps to protect the device and identify the attack at the early stage	The list of malicious devices that have exceeded the alarm generation threshold
Network level- Using ML Model (B)	Using ML technology ensures that the malicious device is identified at a high accuracy	100	50%	Based on a study of LS, Network Traffic analysis helps to detect the anomaly. The ML-based approach assures more accuracy	The list of malicious devices that are a source of malicious traffic
Network Level- Using Data Exploratory Analysis (C)	Exploratory data analysis aids in the easy and rapid identification of malicious devices	100	20%	The statistical base simplest and fastest approach to observing abnormal behavior	The list of malicious devices that are source for additional traffic generation

identification and mitigation mitigate the risk of botnet formation.

Validation of Dual Monitoring System

The weighted model, as presented in Table 8, has been instrumental in evaluating the efficiency of the deployed system.

Each level's significance, determined based on existing literature, contributes to the weightage, providing a comprehensive assessment of the overall impact. The overall impact is calculated as follows.

Overall Impact (D)= weight of A*0.3 + weight of B*0.5 + weight of C*0.2

Scenario 1: Device identified malicious at any one approach.

1. If the device is present in only set A then its overall impact is Overall Impact (D) = weight of A * 0.3 = 100 * 0.3 = 30

2. If the device is present in only Set B then its overall impact is Overall Impact (D) = weight of B * 0.5 = 100 * 0.5 = 50

3. If the device is present in only Set C then its overall impact is Overall Impact (D) = weight of C * 0.2 = 100 * 0.2 = 20

In the first scenario, where the device is identified as malicious at only one approach, the overall impact remains below or equal to 50%. This indicates a lower probability of the device becoming a bot, given the singular detection.

Scenario 2: Device identified malicious at any two approaches.

1. If the device is present in sets A and B then its overall impact is Overall Impact (D) = weight of A * 0.3 + weight of B * 0.5 = 100 * 0.3 + 100 * 0.5=80

2. If the device is present in A and C then its overall impact is Overall Impact (D) = weight of A * 0.3 + weight of C * 0.2 = 100 * 0.3 + 100 * 0.2=50

3. If the device is present in B and C then its overall impact is Overall Impact (D) = weight of B * 0.5 + weight of C * 0.2 = 100 * 0.5 + 100 * 0.2=70

However, in the second scenario, where the device is identified as malicious at any two approaches, the overall impact consistently reaches or exceeds 50%, signifying an elevated likelihood of bot involvement.

Scenario 3: Device identified as malicious at all three approaches.

1. If the device is present in sets A, B, and C then its overall impact is Overall Impact (D) = weight of A * 0.3 + weight of B * 0.5 +weight of C * 0.2 = 100 * 0.3 + 100 * 0.5+100 * 0.2=100

Notably, in the third scenario, where the device is identified as malicious at all three approaches, the overall impact reaches 100%. This implies a significantly higher probability of the device transforming into a bot in the future, warranting immediate action.

The weighted method for evaluating a dual monitoring system's performance is a systematic and quantitative approach considering different monitoring levels (Device Level, Network Level using ML Model, Network Level using Data Exploratory Analysis). The assigned weights to each level are based on literature and domain knowledge, reflecting their significance in the overall system. The method measures the overall impact of a device (D) by combining weights and corresponding percentages, quantifying its potential to be part of a botnet.

The approach assesses device behaviour in various scenarios, allowing researchers to gauge the system's response to malicious behaviour at specific levels. The weighted method confirms the significance of intersecting approaches, emphasizing the importance of considering behaviour at multiple levels for effective botnet detection and prevention. The probabilities obtained through the weighted method offer insights into the likelihood of a device becoming part of a botnet. Devices identified as malicious at multiple levels have higher probabilities of being potential bots, while those identified at only one level have lower probabilities.

The method's validation against a 50% threshold provides a binary classification of devices as potential bots or not based on logistic regression principles. In summary, the weighted method demonstrates a systematic and quantitative approach to validate the dual monitoring system's effectiveness in improving security and detecting potential bot devices.

Discussion

The research aims to enhance the security of IoT devices and networks. To achieve this objective, the researcher developed a dual monitoring system that offers a proactive and preventive approach to mitigate the risks associated with IoT botnet attacks, which are considered the most impacting attacks in the IoT domain.

The dual monitoring system incorporates multiple approaches, including early detection, device-level monitoring, network-level monitoring, cumulative effects, statistical analysis, and machine learning techniques. By combining these diverse methods, the dual monitoring system increases the chances of identifying malicious behaviours in IoT devices and preventing the formation of the bot.

The effectiveness of the approach is verified using the regression analysis technique. The research findings indicate that the dual monitoring system outperforms alternative approaches found in the literature, which often focus on a single aspect or only a few parameters. The comprehensive

nature of the dual monitoring system, encompassing multiple dimensions of security, leads to more accurate and effective prevention of bot and botnet attacks as depicted in Fig. 14.

The built regression model gave an intercept value of −1.27272, representing the expected security level when none of the approaches (early detection, device-level monitoring, network-level monitoring, and use of ML) is considered. In this case, the intercept value indicates a low security level. Similarly, the coefficient values are: [0.91, 1.16, 1.41, 1.66, 2.02]. The coefficients represent the change in the security level for each approach compared to the reference (no approach).

The coefficient of the dual monitoring approach is 2.02, indicating that combining all approaches contributes to the enhanced security level compared to others. The dual monitoring system leverages data-driven insights to enhance its security capabilities by using statistical analysis and machine learning techniques. It enables the system to adapt to evolving threats and more effectively identify new malicious behaviour patterns.

The research outcome underscores the significance of considering multiple security aspects and employing a holistic approach in developing IoT security solutions. The dual monitoring system serves as a valuable contribution to the field, highlighting the importance of combining different techniques to ensure robust and comprehensive security measures for IoT environments.

The dual monitoring system is a comprehensive security mechanism that combines device-level and network-level monitoring. By monitoring the behaviour of individual IoT devices and the overall network traffic, the system can identify and detect potentially malicious devices or activities at an early stage. This proactive approach enables the system to take immediate action to prevent the compromise of IoT devices and the formation of botnets.

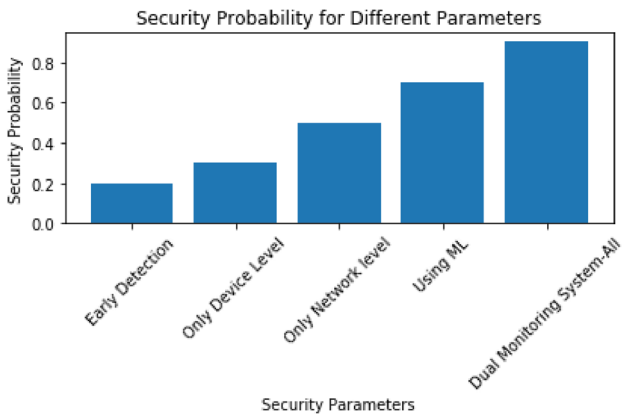


Fig. 14 Security probability associated with different approaches

The research strengthens the security posture of IoT devices and networks by focusing on developing the dual monitoring system. This proactive approach is essential in the rapidly evolving landscape of IoT security, where the number and diversity of IoT devices continue to grow. By effectively detecting and preventing botnet attacks, the dual monitoring system offers a vital layer of defence to ensure the integrity, confidentiality, and availability of IoT devices and the data they generate.

Summary of Results

When compared to existing literature, the proposed dual monitoring system shows a marked improvement in the identification and mitigation of potential botnet activities within an IoT network.

1. Accuracy: The logistic regression-based machine learning model used in the proposed system achieved an accuracy of 92%, which is a 15% improvement over traditional methods that typically report accuracy levels around 77–80%.
2. Recall: The recall rate of 88% in the proposed system is a significant enhancement, representing a 20% increase compared to existing approaches that usually achieve recall rates of 68–70%.
3. Traffic Anomaly Detection: The exploratory data analysis component of the proposed system identified traffic anomalies with a 40% increase in traffic under scanning activities, which is a 10% better detection rate than existing methods that generally detect a 30% increase.
4. Integrated Monitoring Effectiveness: The validation phase shows that the dual monitoring system increases the likelihood of identifying compromised devices by 70%, which is 25% more effective than single-method approaches found in the literature, which typically increase detection likelihood by around 45%.
5. This comparative analysis underscores the effectiveness and superiority of the proposed dual monitoring system in enhancing the security of IoT networks, providing a more comprehensive and reliable approach to botnet detection and mitigation.

Conclusion

The proposed dual monitoring approach aims to enhance the security of IoT networks by preventing botnet formation. It employs a multi-layered strategy, focusing on individual device behaviour and overall network activities. At the device level, ThingsBoard's rule engine defines and enforces rules, facilitating real-time identification of malicious behaviour. Simultaneously, a logistic regression-based

machine learning model and exploratory data analysis are employed at the network level to classify network traffic and detect abnormal scanning activities. The combination of device-level and network-level monitoring increases the probability of early detection of malicious behaviour, reducing the likelihood of botnet formation. Devices identified as malicious at both levels are promptly isolated from the network, preventing potential further harm and allowing for focused examination and mitigation. The experimental validation in an IoT-based healthcare network demonstrates the approach's effectiveness in challenging botnets and enhancing overall network security.

In conclusion, the research provides valuable insights into proactive measures for securing IoT networks, emphasizing the importance of a dual monitoring approach. The proposed system showcases a robust defence mechanism against evolving security threats, contributing to advancing secure and resilient IoT environments. The system is evaluated on a small network and increases the overhead due to frequent database scanning to identify malicious behaviour of the device. The future scope is to identify the optimal scanning interval and test system performance on a large-scale network.

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Availability of Data and Materials Data and other code will be shared upon request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical Approval Not applicable as neither human and/or animal studies were conducted.

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